Ayrton Porto

Mathematician · Formal Methods · Autoformalization

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Research Interests

Autoformalization, formal verification, large language models for mathematics, proof assistants (Lean 4, Coq/Rocq), mathematical logic, algebraic logic, fidelity metrics for machine-generated proofs.

Education

Universidad Nacional del Centro de la Provincia de Buenos Aires (UNICEN) 2020–2025
Licenciatura en Ciencias Matemáticas — Tandil, Argentina

— GPA: **9.41/10.00**. Five-year degree, equivalent to a combined B.S. + M.S.

— Thesis: *Priestley duality for bounded distributive lattices with normal operators*

— Advisors: W. Zuluaga Botero and A. Nagy

Research Experience

AViD Journal — Automated Verification Pipeline 2025–Present
Independent Research Project — Deployed, in active development (avidjournal.com.ar)

— End-to-end pipeline for LLM-assisted autoformalization of mathematical papers in Lean 4, with novelty detection against Mathlib.

— Integrated arXiv and TheoremSearch for paper ingestion; modular Python backend with FastAPI, SSE streaming, and Claude API.

— Presented as a poster at *Escuela de Primavera en Deep Learning* (2025).

— Source: github.com/ayrtonporto/avid-journal

Apart Research — Secure Program Synthesis Fellowship 2026–Present
Fellow Researcher · International collaboration

— Developing **BabelBench**, a cross-paradigm proof-translation benchmark measuring whether LLMs preserve logical meaning when translating theorems across formal systems (Lean 4, Coq, TLA+, Dafny).

— Leading the fidelity-metrics component: metrics checked by real verifiers and Z3 — no LLM judges.

CIN Scholarship — Consejo Interuniversitario Nacional 2023–2025
Undergraduate Research Fellow · UNICEN

— Research in algebraic logic under W. Zuluaga Botero and A. Nagy.

— Studied duality theory for bounded distributive lattices, culminating in the Licenciatura thesis.

Publications & Preprints

Ayrton Porto. "Beyond Correctness: Toward Automated Novelty Verification with Lean 4." Submitted to arXiv (cs.AI), 2026.

Talks & Presentations

An Introduction to Lean 4 — Taller Lean, Nodo Serrano	2026
AViD Journal: LLM-Assisted Autoformalization — Deep Learning Spring School (poster)	2025
Priestley Duality for Bounded Distributive Lattices — Thesis defense, UNICEN	2025
Multivariate Integration via Space-Filling Curves — XII Congreso Mar y Sierras	2023

Teaching Experience

Teaching Assistant — UNICEN 2026–Present
Graduate Teaching Assistant
Courses: Discrete Mathematics, Linear Algebra, Calculus I.

Teaching Assistant — UNICEN 2023–2025
Undergraduate Teaching Assistant
Courses: Discrete Mathematics, Linear Algebra, Calculus I.

Awards & Fellowships

Apart Research — Secure Program Synthesis Fellowship	2026
Santander Scholarship — Academic Merit	2024
CIN Scholarship — Consejo Interuniversitario Nacional	2023–2025
IMPA Summer Program — Rio de Janeiro, Brazil	2023
Course: *Computação, lógica e teoria dos conjuntos.* Grade: A–.	
Honorable Mention — Argentine Mathematical Olympiad (OMA)	2017, 2019

Languages

Spanish (native) · English (professional proficiency) · Portuguese (reading)

Technical Skills

Proof assistants	Lean 4, Coq / Rocq
Programming	Python, C++, SQL
ML / AI	LLM pipelines (Claude API, multi-agent), PyTorch, prompt engineering
Formal methods	Theorem proving, formal verification, model checking (TLA+, Dafny)
Tools	Git, LaTeX, FastAPI, Streamlit, Pandas

References

Available upon request.